HIGH VOLTAGE · ISOLATION

Isolation faults

How the car finds a leak, which codes matter, and what to write on the ticket.

October 2026 · Model S, X, 3, Y, and newer · Match the full alert name, including the letter

SAY IT THIS SIMPLY

The strong electricity that moves the car has to stay inside its own path, like juice in a straw. Isolation means that juice does not touch the metal body. If the wall between them gets a hole — water, a cracked cable, a wet plug — the car stops charging or driving so nobody gets a shock. It is being careful, not stubborn.

How the car finds it

A tester inside the battery computer sends a very small signal from the high-voltage wires toward the body. It is not the driving current. If almost nothing leaks across, the car is fine. If some of that signal sneaks through, the car measures how bad the leak is, remembers whether it is the plus side or the minus side, and sets a warning. It may refuse to drive or charge.

Open contactors or closed

The pack contactors are the big switches between the battery and the rest of the car. Open: only the battery is in the test. If the leak is still there, the hole is inside the pack. Closed: motors, charger, heater, compressor, and orange cables are included. Fine when open and bad when closed means the hole is outside the battery. That one comparison cuts the search in half.

How a technician finishes the hunt

They read the resistance, which side, and whether the pack alone is guilty. Then they disconnect high-voltage parts one at a time and test again: rear drive unit, front drive unit, heater, A/C compressor, charge port. The moment the leak disappears, the last part unplugged is the failed part. If the leak is inside the pack, they look for water, old coolant, corrosion, or damaged insulation. Eyes matter too: cracked orange cable, a soggy charge port, chewed wires, green crust on a connector.

YOU DO NOT DO THIS PART

Do not unplug orange connectors, run an insulation tester, or clear isolation counters to get the car out the door. High voltage can kill. Your job is the exact code, the story, and the right bay: pack, external high voltage, or drive unit.

Read the code before you say battery

The same number with w, a, or f is one family at different severity, not a different part. w may still drive. a limits the car. f has already shut something down. Do not tell a w customer they are fine.

Code	When	What it means	Screen and ticket
BMS_a123 / w123 / f123 SW_Internal_Isolation	Contactors open	Leak is in the circuit with the rest of the car disconnected. Treat it as the HV battery until a tech proves otherwise.	Often Unable to charge. Do not quote a cable or a compressor.
BMS_a035 / w035 / f035 SW_Isolation	Contactors closed	Below the hard limit. Tesla's own text says external or internal. A BMS code is not a battery verdict.	Vehicle may not restart. Need an open-contactor reading before anyone picks a part.
BMS_a151 / w151 / f151 SW_external_isolation	Contactors closed	Pointed outside the pack: compressor, PTC or coolant heater, charge port, charger / DC-DC, drive unit, orange harness.	Vehicle may not restart. Service is required. Book external HV, not a pack quote.
BMS_a142 SW_Isolation_Degradation	Contactors just closed	Under the warning line, still above the stop-the-car line. The quiet early leak. Often stamped at the end of a Supercharge. One hit can fade. A repeat is real.	Usually no message. Ask for the kΩ number and whether every DC session does it.

Code	When	What it means	Screen and ticket
BMS_a027 / w027 / f027 Repeated isolation failure in drive	Many drive cycles	This is the escalation, not the first blip. Isolation already failed across a lot of drives. Capture a companion BMS_a170. Do not guess that code.	Power limited — OK to drive, and may not restart after this drive. Same day. Tow if it already will not restart.
BMS_w172 SW_Drive_Iso_Warning often with BMS_u008	Driving, power reduced	A drive-unit leak wearing a battery code. On 2012–2020 S/X the usual cause is coolant past the rear large-drive-unit rotor seal.	Ask about coolant top-offs. Write “rule out rear LDU before the pack.”
BMS_a037 SW_Flood_Port_Open	Moisture path	The pack drain is open. Not the ohms reading itself. It is the door the water used. More common on 3/Y and 2021+ S/X. Older S/X 85 packs leak through breather valves and show up as the a035 family instead.	Ask flood, deep water, underbody wash, and the date. Pair it with the isolation code beside it.

Clear condition on these alerts is the same idea every time: isolation returns above the threshold on a later test. The car retests at power-up. Clearing counters is not a repair.

Traps that wreck the story

- a035 does not mean “external only,” and it does not mean “battery.” The name is SW_Isolation. Open-versus-closed decides.
- a123 with the contactors open is the pack until someone proves it is not.
- w172 is a motor. Classic S/X, rear large drive unit, coolant loss with no puddle under the car.
- a142 is the quiet one. No screen message. Ignore the repeats and it grows into a027.
- a027 means it already failed a lot. “It still drives” is not an all-clear. The next Park may be the one that sticks.
- A dead 12-volt beside an isolation alert is usually the victim. Contactors never close, so the DC-DC never feeds the small battery. Note both. Do not replace the 12-volt and release the car.
- “We cleared it” is not a fix. If the leak is still there, the alert is back before they leave the lot.

These are not isolation

Code	It is actually
BMS_a066, then a064, then a079	Cell imbalance. Range cut, sometimes near half. No chassis leak.
BMS_a017	A brick went over-voltage.
BMS_a166, or older BMS_w152 / w153	Welded contactor. Different safety problem.
BMS_f038 / w026	Stuck contactor, mostly older S/X.
BMS_w107 / f107	Voltage-sense harness tabs. Not insulation.

What their sentence is telling you

They say	Think	Ask
May not restart	a035, a151, or a027. Urgent.	Are they still driving it, or is it already parked?
Will not charge, still drives	a123, or water in the charge port.	Home AC, Supercharger, or both?
Only when I unplug a Supercharger	a142 degradation.	Screenshot of the kΩ, and does every DC session do it?
Only with heat or A/C on	PTC heater on older cars, or the compressor.	Does it start during precondition?
Only after rain or a car wash	Charge port or a harness. Not a pack quote yet.	Does the charge-port door latch?
Old S/X, coolant goes down, power reduced	w172 / rear LDU.	Any sweet smell, any top-offs, any isolation alert yet?
Flood, deep puddle, wet carpet	a037 plus an isolation code.	What date did the water happen?

Put this on the ticket

- Exact alert names, letter included (w / a / f), plus anything cleared in the last 30 days.
- Isolation resistance with contactors open and closed, and the timestamp.
- When it sets: key-on, after a drive, end of DC charge, HVAC on, rain.
- AC charge, DC charge, or both. Which site, if DC.
- Coolant loss, sweet smell, underbody wash, flood, body shop, or a replaced orange cable.
- Model and year. On 2012–2020 S/X, the first line is “rule out the rear LDU before the pack.”
- The customer’s words: “may not restart,” “unable to charge,” or “no message.”

NUMBERS YOU DO NOT QUOTE

Healthy isolation is usually several thousand kΩ. A degradation warning is often under about 1,000 kΩ. A hard fault is lower. A legacy Model S bulletin treated open-contactor isolation under about 1,800 kΩ as a pack problem. Those are the scale techs see, not a spec. Current limits live in the procedure for that VIN. Insulation-test voltage is not the same for every part: PTC heaters and compressors are tested much lower than a charge port. “Megger everything at 1,000 V” can destroy a good compressor.

PRIORITY

a123, f035, f151, or an a027 that already will not restart: tow. Do not ask them to try charging again. a035 or a151 “may not restart”: same day, and tell them the next Park may stick. A single a142 with no message after one Supercharge: note it and watch. Do not scare them into a pack quote.